

X chromosome inactivation in the absence of Dicer.

PMID: 19164542 | Indexed in PubMed

Dicer is central to the RNA interference (RNAi) pathway, because it is required for processing of double-stranded RNA (dsRNA) precursors into small RNA effector molecules. In principle, any long dsRNA could serve as a substrate for Dicer. The X inactive specific transcript (Xist) is an untranslated RNA that is required for dosage compensation in mammals. It coats and silences 1 of the 2 X chromosomes in female cells and initiates a chromosomewide change in chromatin structure that includes the recruitment of Polycomb proteins, but it is largely unknown how Xist RNA mediates these processes. To investigate a potential link between the RNAi pathway and X inactivation, we generated and analyzed Dicer-deficient embryonic stem (ES) cells. In the absence of Dicer, coating by Xist RNA, initiation of silencing, and recruitment of Polycomb proteins occur normally. Dicer ablation had modest effects on the steady-state levels of spliced Xist RNA. Together our data indicate that the RNAi machinery is not essential for the initiation of X inactivation.

Ectopic Splicing Disturbs the Function of Xist RNA to Establish the Stable Heterochromatin State.

PMID: 34722536 | Indexed in PubMed

Non-coding Xist RNA plays an essential role in X chromosome inactivation (XCI) in female mammals. It coats the X chromosome in cis and mediates the recruitment of many proteins involved in gene silencing and heterochromatinization. The molecular basis of how Xist RNA initiates chromosomal silencing and what proteins participate in this process has been extensively studied and elucidated. Its involvement in the establishment and maintenance of the X-inactivated state is, however, less understood. The Xist IVS allele we previously reported is peculiar in that it can initiate XCI but fails to establish the inactive state that is stably maintained and, therefore, may provide an opportunity to explore how Xist RNA contributes to establish a robust heterochromatin state. Here we demonstrate that ectopic splicing taking place to produce Xist IVS RNA disturbs its function to properly establish stable XCI state. This finding warrants the potential of Xist IVS RNA to provide further insight into our understanding of how Xist RNA contributes to establish sustainable heterochromatin.

Spen links RNA-mediated endogenous retrovirus silencing and X chromosome inactivation.

PMID: 32379046 | Indexed in PubMed

The Xist lncRNA mediates X chromosome inactivation (XCI). Here we show that Spen, an Xist-binding repressor protein essential for XCI, binds to ancient retroviral RNA, performing a surveillance role to recruit chromatin silencing machinery to these parasitic loci. Spen loss activates a subset of endogenous retroviral (ERV) elements in mouse embryonic stem cells, with gain of chromatin accessibility, active histone modifications, and ERV RNA transcription. Spen binds directly to ERV RNAs that show structural similarity to the A-repeat of Xist, a region critical for Xist-mediated gene silencing. ERV RNA and Xist A-repeat bind the RRM domains of Spen in a competitive manner. Insertion of an ERV into an A-repeat deficient Xist rescues binding of Xist RNA to Spen and results in strictly local gene silencing in cis. These

results suggest that Xist may coopt transposable element RNA-protein interactions to repurpose powerful antiviral chromatin silencing machinery for sex chromosome dosage compensation.

Xist function: bridging chromatin and stem cells.

PMID: 17681633 | Indexed in PubMed

In mammals, dosage compensation is achieved by transcriptional silencing of one of the two female X chromosomes. X inactivation is dynamically regulated in development. The non-coding Xist RNA localizes to the inactive X, initiates gene repression in the early embryo, and later stabilizes the inactive state. Different functions of Xist are observed depending on which epigenetic regulatory pathways are active in a given cell. Because Xist has evolved recently, with the origin of placental mammals, the underlying pathways are also important in regulating developmental control genes. This review emphasizes the opportunity that Xist provides to functionally define epigenetic transitions in development, to understand cell identity, pluripotency and stem cell differentiation.

Cancer progenitors and epigenetic contexts: an Xisting connection.

PMID: 19923898 | Indexed in PubMed

In mammals, silencing of one of the two X chromosomes is necessary to achieve dosage compensation. The non coding RNA Xist triggers X inactivation. Gene silencing by Xist is only possible in certain developmental contexts that only exist in cells of the early embryo and specific hematopoietic progenitors. Critical silencing factors may only be present in these contexts giving an explanation of why Xist is not operative outside these contexts. It has been demonstrated that Xist is functional in tumor cells, where SATB1 was identified as the first silencing factor for Xist mediated chromosome silencing.

A lifelong duty: how Xist maintains the inactive X chromosome.

PMID: 35717799 | Indexed in PubMed

Female eutherians transcriptionally silence one X chromosome to balance gene dosage between the sexes. X-chromosome inactivation (XCI) is initiated by the lncRNA Xist, which assembles many proteins within the inactive X chromosome (Xi) to trigger gene silencing and heterochromatin formation. It is well established that gene silencing on the Xi is maintained through repressive epigenetic processes, including histone deacetylation and DNA methylation. Recent studies revealed a new mechanism where RNA-binding proteins that interact directly with the RNA contribute to the maintenance of Xist localization and gene silencing. In addition, a surprising plasticity of the Xi was uncovered with many genes becoming upregulated upon experimental deletion of Xist. Intriguingly, immune cells normally lose Xist from the Xi, suggesting that this Xist dependence is utilized in vivo to dynamically regulate gene expression from the Xi. These new studies expose fundamental regulatory mechanisms for the chromatin association of RNAs, highlight the need for studying the maintenance of XCI and Xist localization in a gene- and cell-type-specific manner, and are likely to have clinical impact.

CRISPR/Cas9-mediated modulation of splicing efficiency reveals short splicing isoform of Xist RNA is sufficient to induce X-chromosome inactivation.

PMID: 29237010 | Indexed in PubMed

Alternative splicing of mRNA precursors results in multiple protein variants from a single gene and is critical for diverse cellular processes and development. Xist encodes a long noncoding RNA which is a central player to induce X-chromosome inactivation in female mammals and has two major splicing variants: long and short isoforms of Xist RNA. Although a differentiation-specific and a female-specific expression of Xist isoforms have been reported, the functional role of each Xist RNA isoform is largely unexplored. Using CRISPR/Cas9-mediated targeted modification of the 5' splice site in Xist intron 7, we create mutant female ES cell lines which dominantly express the long- or short-splicing isoform of Xist RNA from the inactive X-chromosome (Xi) upon differentiation. Successful execution of CRISPR/Cas-based splicing modulation indicates that our CRISPR/Cas-based targeted modification of splicing sites is a useful approach to study specific isoforms of a transcript generated by alternative splicing. Upon differentiation of splicing-mutant Xist female ES cells, we find that both long and short Xist isoforms can induce X-chromosome inactivation normally during ES cell differentiation, suggesting that the short splicing isoform of Xist RNA is sufficient to induce X-chromosome inactivation.

Xist-mediated silencing requires additive functions of SPEN and Polycomb together with differentiation-dependent recruitment of SmcHD1.

PMID: 35584662 | Indexed in PubMed

X chromosome inactivation (XCI) is mediated by the non-coding RNA Xist, which directs chromatin modification and gene silencing in cis. The RNA binding protein SPEN and associated corepressors have a central role in Xist-mediated gene silencing. Other silencing factors, notably the Polycomb system, have been reported to function downstream of SPEN. In recent work, we found that SPEN has an additional role in correct localization of Xist RNA in cis, indicating that its contribution to chromatin-mediated gene silencing needs to be reappraised. Making use of a SPEN separation-of-function mutation, we show that SPEN and Polycomb pathways, in fact, function in parallel to establish gene silencing. We also find that differentiation-dependent recruitment of the chromosomal protein SmcHD1 is required for silencing many X-linked genes. Our results provide important insights into the mechanism of X inactivation and the coordination of chromatin-based gene regulation with cellular differentiation and development.

Integrated analysis of Xist upregulation and X-chromosome inactivation with single-cell and single-allele resolution.

PMID: 34131144 | Indexed in PubMed

To ensure dosage compensation between the sexes, one randomly chosen X chromosome is silenced in each female cell in the process of X-chromosome inactivation (XCI). XCI is initiated during early development through upregulation of the long non-coding RNA Xist, which mediates chromosome-wide gene silencing. Cell differentiation, Xist upregulation and gene silencing are thought to be coupled at

multiple levels to ensure inactivation of exactly one out of two X chromosomes. Here we perform an integrated analysis of all three processes through allele-specific single-cell RNA-sequencing. Specifically, we assess the onset of random XCI in differentiating mouse embryonic stem cells, and develop dedicated analysis approaches. By exploiting the inter-cellular heterogeneity of XCI onset, we identify putative Xist regulators. Moreover, we show that transient Xist upregulation from both X chromosomes results in biallelic gene silencing right before transitioning to the monoallelic state, confirming a prediction of the stochastic model of XCI. Finally, we show that genetic variation modulates the XCI process at multiple levels, providing a potential explanation for the long-known X-controlling element (Xce) effect, which leads to preferential inactivation of a specific X chromosome in inter-strain crosses. We thus draw a detailed picture of the different levels of regulation that govern the initiation of XCI. The experimental and computational strategies we have developed here will allow us to profile random XCI in more physiological contexts, including primary human cells in vivo.

Epigenomic landscape of the human dorsal root ganglion: sex differences and transcriptional regulation of nociceptive genes.

PMID: 38586055 | Indexed in PubMed

Gene expression is influenced by chromatin architecture via controlled access of regulatory factors to DNA. To better understand gene regulation in the human dorsal root ganglion (hDRG) we used bulk and spatial transposase-accessible chromatin technology followed by sequencing (ATAC-seq). Using bulk ATAC-seq, we detected that in females diverse differentially accessible chromatin regions (DARs) mapped to the X chromosome and in males to autosomal genes. EGR1/3 and SP1/4 transcription factor binding motifs were abundant within DARs in females, and JUN, FOS and other AP-1 factors in males. To dissect the open chromatin profile in hDRG neurons, we used spatial ATAC-seq. The neuron cluster showed higher chromatin accessibility in GABAergic, glutamatergic, and interferon-related genes in females, and in Ca²⁺-signaling-related genes in males. Sex differences in transcription factor binding sites in neuron-proximal barcodes were consistent with the trends observed in bulk ATAC-seq data. We validated that EGR1 expression is biased to female hDRG compared to male. Strikingly, XIST, the long-noncoding RNA responsible for X inactivation, hybridization signal was found to be highly dispersed in the female neuronal but not non-neuronal nuclei suggesting weak X inactivation in female hDRG neurons. Our findings point to baseline epigenomic sex differences in the hDRG that likely underlie divergent transcriptional responses that determine mechanistic sex differences in pain.